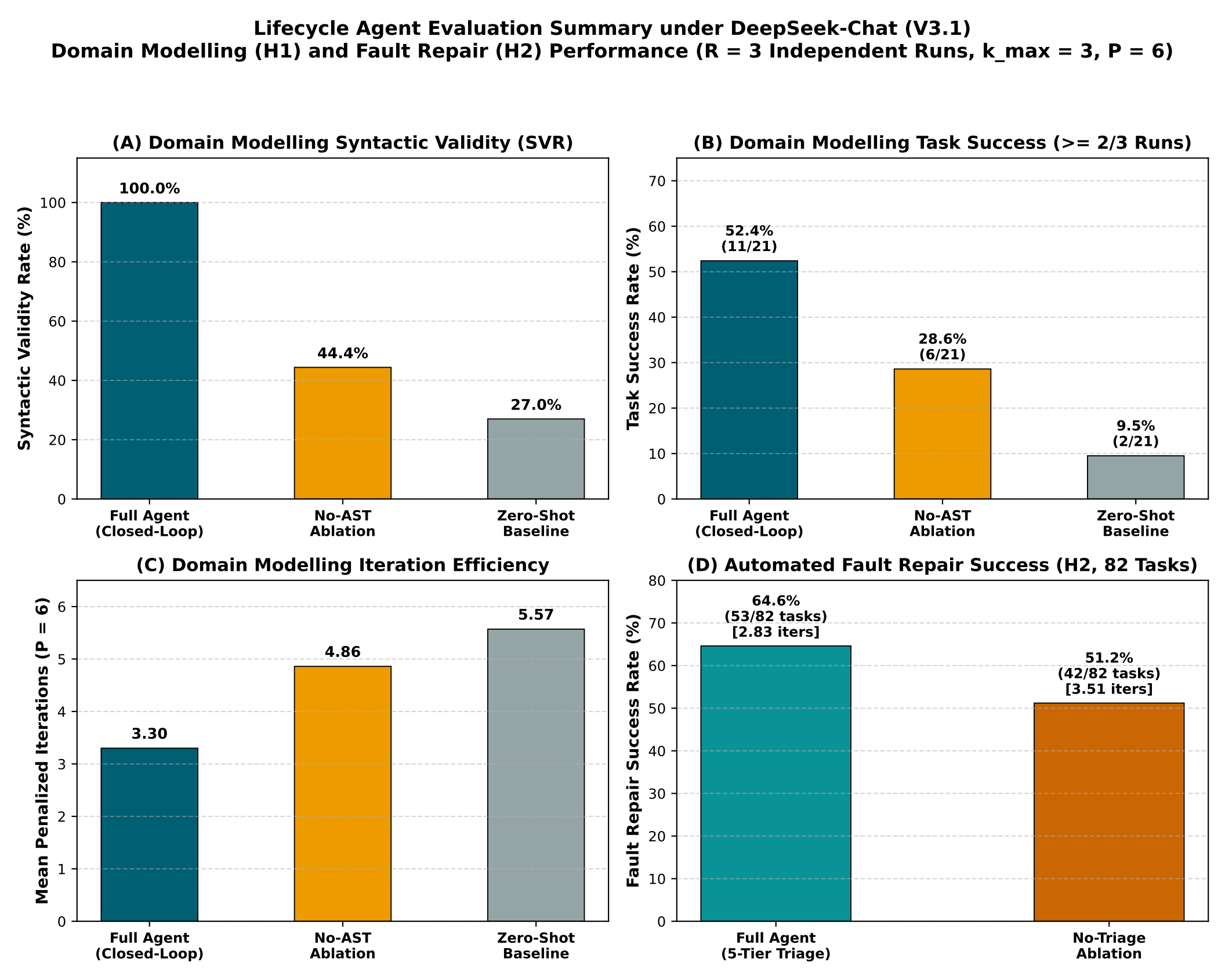
Lifecycle Agent Evaluation Summary under DeepSeek-Chat (V3.1)
Domain Modelling (H1) and Fault Repair (H2) Performance (R = 3 Independent Runs, k_max = 3, P = 6)
(A) Domain Modelling Syntactic Validity (SVR)
Syntactic Validity Rate (%)
100.0%
44.4%
27.0%
Full Agent (Closed-Loop)
No-AST Ablation
Zero-Shot Baseline
(B) Domain Modelling Task Success (>= 2/3 Runs)
Task Success Rate (%)
52.4% (11/21)
28.6% (6/21)
9.5% (2/21)
Full Agent (Closed-Loop)
No-AST Ablation
Zero-Shot Baseline
(C) Domain Modelling Iteration Efficiency
Mean Penalized Iterations (P = 6)
3.30
4.86
5.57
Full Agent (Closed-Loop)
No-AST Ablation
Zero-Shot Baseline
(D) Automated Fault Repair Success (H2, 82 Tasks)
Fault Repair Success Rate (%)
64.6% (53/82 tasks) [2.83 iters]
51.2% (42/82 tasks) [3.51 iters]
Full Agent (5-Tier Triage)
No-Triage Ablation